Exercises

HUJI 67594 · updated weekly

How to use. Each week has its questions first and its solutions on a separate page, so you can work without seeing the answers. Every solution gives the *idea* (which tool and why), numbered steps with all numbers, a boxed *answer*, and a *watch out* line where people often slip. Difficulty: ● ○ ○ direct use of a formula, ● ● ○ several steps, ● ● ● exam level. Sources: `[KR 6.P11]` = Kurose & Ross, 8th ed., chapter 6, problem 11 (reworded); `[R2]`, `[T1]` = a variant of a recitation or tutorial question with new numbers; `[own]` = written for this sheet. The formulas are in `formula_sheet.pdf`; the explanations are in `weekNN.pdf`.

Contents. Week 01: delay, switching, probability (14 questions). Week 02: layering, ALOHA, CSMA/CD, WiFi (18 questions).

WEEK 01 — Lecture 1: Introduction · Tutorial 1: Probability

Delay and switching

Q1.1 ● ○ ○ `[KR 1.P6]`

Hosts A and B are m metres apart, joined by one link of rate R bit/s with propagation speed s. A sends one packet of L bits starting at $t = 0$. (a) Write d_{prop}, d_{trans} and the end-to-end delay (no processing or queuing). (b) At $t = d_{\text{trans}}$, where is the last bit? (c) If $d_{\text{prop}} > d_{\text{trans}}$, where is the first bit at $t = d_{\text{trans}}$? (d) With $s = 2.5 \times 10^8$ m/s, $L = 1500$ bytes and $R = 10$ Mbps, for which m is $d_{\text{prop}} = d_{\text{trans}}$?

Q1.2 ● ● ○ `[KR 1.P10]`

A 1500-byte packet crosses three links joined by two switches. Every link has $R = 2.5$ Mbps and $s = 2.5 \times 10^8$ m/s; the links are 5000, 4000 and 1000 km long. Each switch adds a processing delay of 3 ms. There is no queuing. Find the end-to-end delay.

Q1.3 ● ○ ○ `[KR 1.P12]`

A packet arrives at a router. On its outgoing link one packet is half transmitted and four more are waiting. All packets are 1500 bytes and the link is 2.5 Mbps. (a) What is the queuing delay of the new packet? (b) Give the general formula when x bits of the current packet are already sent and n packets wait.

Q1.4 ● ● ○ `[KR 1.P31]`

A message of 10^6 bits goes from a source through two store-and-forward switches to a destination (three links, 5 Mbps each). Ignore propagation, queuing and processing. (a) How long does it take without segmentation? (b) Now it is cut into 100 packets of 10 000 bits. When does the first packet reach the first switch? When is the second packet fully received there? (c) How long does the whole message take now? Why is it faster?

Q1.5 ● ○ ○ `[own]`

A 1 000 000-bit file is sent over a circuit-switched path. Every link is a 2 Mbps TDM link with 20 slots, and setting up the circuit takes 300 ms. How long until the whole file has arrived (ignore propagation)? Does it depend on the number of links?

Q1.6 ● ○ ○ `[own]`

An application has peak rate 1 Mbps and average rate 50 kbps. (a) What is its utilization under circuit switching? (b) How many such applications fit on a 10 Mbps link with circuit switching? (c) How many would fit if each needed only its average rate?

Q1.7 ●○○ [own]

Packets of 1000 bits arrive at a 1 Mbps link at $a = 900$ packets/s on average. (a) What is the traffic intensity? (b) Up to which arrival rate is the queue stable?

Q1.8 ●●○ [KR 1.P8]

Users share a 10 Mbps link. Each needs 200 kbps when active and is active 10% of the time, independently. (a) How many users does circuit switching support? (b) With packet switching and 120 users, what is the probability that exactly n users are active? (c) What is the probability that 51 or more are active? What does that mean for packet switching?

Q1.9 ●●○ [KR 1.P9]

Same as Q1.8 but with a 1 Gbps link and users of 100 kbps, each active with probability $p = 0.1$. (a) What is N, the number of users under circuit switching? (b) With packet switching and M users, write the probability that more than N are active.

Probability

Q1.10 ●●○ [T1 / lecture Example II]

Good packets survive t seconds with probability e^{-t}, bad ones with probability e^{-1000t}. A packet is good with probability 0.2. (a) What is the probability that a packet survives 2 ms? (b) It survived 2 ms: what is the probability that it is good?

Q1.11 ●○○ [KR 1.P22]

A path has 5 links, each losing a packet independently with probability 0.05. The server resends a lost packet end to end. (a) Probability that one transmission arrives? (b) Expected number of *re*transmissions?

Q1.12 ●●● [T1 / lecture Example III]

A chain of $n = 4$ links, each failing independently with probability $p = 0.2$. (a) With end-to-end retransmission, what is the expected number of attempts by S, and the expected number of *link* transmissions? (b) With hop-by-hop retransmission, what is the expected number of link transmissions? Which scheme is cheaper?

Q1.13 ●●○ [own]

Packets arrive as a Poisson process with rate $\lambda = 3$ per second. (a) Probability of no packet in 1 s? (b) Probability of exactly 2 packets in 0.5 s? (c) You have already waited 2 s without a packet. What is the expected remaining wait? Why?

Q1.14 ●○○ [T1]

In each interval of length T the number of packets sent is Pois(gT). (a) What is the probability of exactly one packet? (b) For which gT is it largest, and what is the largest value?

WEEK 01 · SOLUTIONS

Q1.1

Idea. Two different delays. Transmission = the time to push all the bits out of A (depends on L and R). Propagation = the time one bit needs to travel the wire (depends on m and s).

(a) $d_{\text{prop}} = m/s$ and $d_{\text{trans}} = L/R$. The last bit leaves A at $t = L/R$ and needs m/s more to arrive, so the end-to-end delay is $L/R + m/s$.

(b) At $t = d_{\text{trans}}$, A has just pushed out its last bit: the last bit is *at* A, just leaving.

(c) The first bit left at $t = 0$ and has travelled for d_{trans} seconds, i.e. $s \cdot d_{\text{trans}}$ metres. Since $d_{\text{prop}} > d_{\text{trans}}$, this is less than m: the first bit is still on the wire and has not reached B.

(d) Set $m/s = L/R$, so $m = sL/R$. $L = 1500 \times 8 = 12\,000$ bits, $L/R = 12\,000/10^7 = 1.2\,\text{ms}$, $m = 2.5 \times 10^8 \times 1.2 \times 10^{-3} = 3 \times 10^5\,\text{m}$.

Answer. (a) $L/R + m/s$; (b) at A, leaving; (c) $s\,d_{\text{trans}}$ from A, still on the wire; (d) $m = 300\,\text{km}$.

Watch out: bytes → bits (×8) before dividing by R.

Q1.2

Idea. Store-and-forward: each switch waits for the *whole* packet before sending it on, so the transmission time is paid on every link. Add the four kinds of delay.

1. Transmission: $L = 1500 \times 8 = 12\,000$ bits; per link $12\,000/(2.5 \times 10^6) = 4.8\,\text{ms}$; three links: $14.4\,\text{ms}$.
2. Propagation: $5 \times 10^6/2.5 \times 10^8 = 20\,\text{ms}$, $4 \times 10^6/2.5 \times 10^8 = 16\,\text{ms}$, $10^6/2.5 \times 10^8 = 4\,\text{ms}$; total $40\,\text{ms}$.
3. Processing: only at the 2 switches: $2 \times 3 = 6\,\text{ms}$.
4. Queuing: 0 (given).

Answer. $14.4 + 40 + 6 = 60.4\,\text{ms}$.

Watch out: km → m; processing happens at 2 switches, not 3.

Q1.3

Idea. The new packet waits until every bit ahead of it has been transmitted.

1. One packet takes $L/R = 12\,000/(2.5 \times 10^6) = 4.8\,\text{ms}$.
2. Ahead of it: the remaining half of the current packet ($2.4\,\text{ms}$) and 4 full packets ($4 \times 4.8 = 19.2\,\text{ms}$).
3. General case: the current packet still has $L - x$ bits to send, and n packets hold nL bits.

Answer. (a) $2.4 + 19.2 = 21.6\,\text{ms}$; (b) $\dfrac{nL + (L - x)}{R}$.

Q1.4

Idea. Without segmentation every link must receive the whole message before the next link can start. With packets, the links work at the same time, like a pipeline.

(a) The message takes $10^6/(5 \times 10^6) = 0.2\,\text{s}$ on one link. It is stored at each switch, so the 3 links are used one after the other: $3 \times 0.2 = 0.6\,\text{s}$.

(b) One packet takes $10^4/(5 \times 10^6) = 2\,\text{ms}$ per link. Packet 1 reaches switch 1 at $t = 2\,\text{ms}$. From 2 to 4 ms the source sends packet 2 (while switch 1 forwards packet 1), so packet 2 is fully at switch 1 at $t = 4\,\text{ms}$.

(c) Packet 1 needs $3 \times 2 = 6\,\text{ms}$ to reach the destination. After that, one packet arrives every 2 ms, so the other 99 arrive $99 \times 2 = 198\,\text{ms}$ later. Total $6 + 198 = 204\,\text{ms}$. This is the formula $(N + K - 1)L/R = (3 + 100 - 1) \times 2\,\text{ms}$.

Answer. (a) 0.6 s; (b) 2 ms and 4 ms; (c) 204 ms, about 3 times faster, because all three links are busy at the same time.

Q1.5

Idea. TDM gives the circuit a fixed share of the link rate. The setup is paid once, and then the bits stream through.

1. The circuit's rate is $2\,\text{Mbps}/20 = 100\,\text{kbps}$.
2. Transmission: $10^6/10^5 = 10\,\text{s}$.
3. Add the setup: $10 + 0.3 = 10.3\,\text{s}$.

Answer. 10.3 s. It does not depend on the number of links: the switches do not store the file, they pass the bits straight through the reserved path.

Q1.6

Idea. A circuit must reserve the *peak* rate. Utilization = what is used on average ÷ what is reserved.

(a) $A/P = 50\,\text{kbps}/1000\,\text{kbps} = 5\%$.

(b) Each reserves 1 Mbps: $10/1 = 10$ applications.

(c) Each needs only 50 kbps: $10\,000/50 = 200$ applications.

Answer. 5%; 10; 200. Packet switching (statistical multiplexing) gets close to (c), because the applications rarely peak at the same time.

Q1.7

Idea. Traffic intensity $I = La/R$ = bits arriving per second ÷ bits the link can send per second.

(a) Arriving: $1000 \times 900 = 9 \times 10^5$ bit/s. $I = 9 \times 10^5/10^6 = 0.9$.

(b) Stable needs $La < R$: $a < 10^6/1000 = 1000$ packets/s.

Answer. (a) 0.9; (b) $a < 1000$ packets/s.

Watch out: $I < 1$ only means the queue does not grow forever. At $I = 0.9$ the queuing delay is already large.

Q1.8

Idea. Circuit switching: divide the link rate. Packet switching: count the active users, which is a binomial random variable.

(a) $10\,\text{Mbps}/200\,\text{kbps} = 50$ users.

(b) Each of the 120 users is active independently with probability 0.1, so $X \sim \text{Bin}(120, 0.1)$ and $P(X = n) = \binom{120}{n} 0.1^n 0.9^{120-n}$.

(c) The link holds 50 active users, so overload means $X \geq 51$: $P(X \geq 51) = \sum_{n=51}^{120} \binom{120}{n} 0.1^n 0.9^{120-n}$. Sense check: the mean is $np = 12$ and the standard deviation $\sqrt{np(1-p)} = \sqrt{10.8} \approx 3.3$, so 51 is $(51 - 12)/3.3 \approx 12$ standard deviations above the mean. Numerically the sum is about 2×10^{-20}.

Answer. (a) 50; (b) $\binom{120}{n} 0.1^n 0.9^{120-n}$; (c) $\approx 2 \times 10^{-20}$: packet switching serves 120 users where circuits serve 50, and is practically never overloaded.

Q1.9

(a) $N = 10^9/10^5 = 10\,000$ users.

(b) Active users $X \sim \text{Bin}(M, p)$. "More than N" means $X \geq N + 1$.

Answer. (a) 10 000; (b) $P(X > N) = \sum_{k=N+1}^{M} \binom{M}{k} p^k (1-p)^{M-k} = 1 - \sum_{k=0}^{N} \binom{M}{k} p^k (1-p)^{M-k}$.

Q1.10

Idea. (a) We do not know the packet's type, so split on it (total probability). (b) We know the result and ask about the cause, so reverse the condition (Bayes).

Let A = "survives $t = 2\,\text{ms}$" and B = "good". Given: $P(B) = 0.2$, $P(A \mid B) = e^{-0.002} = 0.998$, $P(A \mid \overline{B}) = e^{-1000 \cdot 0.002} = e^{-2} = 0.135$.

(a) $P(A) = P(A \mid B)P(B) + P(A \mid \overline{B})P(\overline{B}) = 0.998 \cdot 0.2 + 0.135 \cdot 0.8 = 0.1996 + 0.1083 = 0.308.$

(b) $P(B \mid A) = \dfrac{P(A \mid B)P(B)}{P(A)} = \dfrac{0.1996}{0.308} = 0.648.$

Answer. (a) 0.308; (b) 0.648. Surviving raised the chance of "good" from 0.2 to 0.65, because bad packets rarely survive even 2 ms.

Q1.11

(a) The packet must survive all 5 links; they are independent, so multiply: $0.95^5 = 0.774$.

(b) Each attempt succeeds independently with probability 0.774, so the number of attempts is Geo(0.774) with mean $1/0.774 = 1.292$. The first attempt is not a retransmission: $1.292 - 1 = 0.292$.

Answer. (a) 0.774; (b) 0.29 retransmissions on average.

Watch out: "retransmissions" = attempts -1.

Q1.12

Idea. End to end: a loss anywhere repeats the whole path. Hop by hop: a loss repeats only that link. Count link transmissions in both cases.

(a) One attempt succeeds with probability $0.8^4 = 0.4096$, so the attempts are Geo(0.4096): mean $1/0.4096 = 2.44$. An attempt always uses link 1; it uses link 2 only if link 1 worked (0.8), link 3 only if links 1–2 worked (0.64), link 4 with probability 0.512. By linearity of expectation, one attempt uses $1 + 0.8 + 0.64 + 0.512 = 2.952$ links on average. Total: $2.44 \times 2.952 \approx 7.2$ link transmissions.

(b) Each hop is retried until it works: Geo(0.8), mean $1/0.8 = 1.25$ sends. Four hops: $4 \times 1.25 = 5$.

Answer. (a) 2.44 attempts, ≈ 7.2 link transmissions; (b) 5. Hop by hop is cheaper.

Q1.13

Idea. In a Poisson process with rate λ, the count in a window of length t is Pois(λt), and the gaps are Exp(λ), which is memoryless.

(a) 1 s: Pois(3). $P(0) = e^{-3} = 0.050$.

(b) 0.5 s: Pois(1.5). $P(2) = e^{-1.5}\,\dfrac{1.5^2}{2!} = 0.2231 \times 1.125 = 0.251.$

(c) Memorylessness: the remaining wait is again Exp(3), with mean 1/3 s, whatever the time already waited.

Answer. (a) 0.050; (b) 0.251; (c) 1/3 s.

Watch out: Scale the mean with the window: λt, not λ.

Q1.14

(a) $P(X = 1) = \dfrac{(gT)^1}{1!}e^{-gT} = gT\,e^{-gT}.$

(b) Write $x = gT$ and $f(x) = xe^{-x}$. Then $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$, which is zero at $x = 1$; f rises before and falls after, so it is a maximum. $f(1) = e^{-1}$.

Answer. (a) $gT\,e^{-gT}$; (b) largest at $gT = 1$, value $1/e \approx 0.368$. This is slotted ALOHA's best goodput (week 2).

WEEK 02 Lecture 2: Layering & Broadcast · Recitation 2: ALOHA

Layering

Q2.1 ●○○ [own]

An application sends 200 bytes over TCP/IP/Ethernet. (a) What fraction of the headers-plus-data is useful data? (b) What if you also count Ethernet's 4-byte CRC and 8-byte preamble?

Q2.2 ●○○ [R2]

Give the OSI layer of: HTTP, TLS encryption, TCP acknowledgements, IP routing, CSMA/CD, the optical fibre. Which of them are not a separate layer in the Internet's 5-layer stack?

Channel partitioning and ALOHA

Q2.3 ●○○ [own]

A 40 Mbps channel is shared by TDMA among 8 stations; only 3 have data. (a) Rate of each busy station? (b) How much capacity is wasted? (c) Which wish-list property fails?

Q2.4 ●●○ [KR 6.P8, P9]

(a) For slotted ALOHA with N nodes, find the p that maximizes $Np(1-p)^{N-1}$ and the limit of the maximum as $N \to \infty$. (b) Do the same for pure ALOHA, $Np(1-p)^{2(N-1)}$.

Q2.5 ●●○ [KR 6.P11]

Four nodes A, B, C, D use slotted ALOHA; each always has frames and sends in each slot with probability p. Slots are numbered $1, 2, \ldots$ (a) Probability that A succeeds for the first time in slot 4? (b) Probability that some node succeeds in slot 5? (c) Probability that the first success (by anyone) is in slot 4? (d) Efficiency of the system? Give all answers as formulas and as numbers for $p = 0.25$.

Q2.6 ●●● [KR 6.P10]

Two nodes A and B use slotted ALOHA with probabilities p_A and p_B. (a) Give A's average throughput (fraction of slots A wins) and the total efficiency. (b) If $p_A = 2p_B$, is A's throughput twice B's? If not, which p_A makes it exactly twice? (c) With N nodes, A using $2p$ and all others p: the throughput of A and of any other node?

Q2.7 ●●○ [R2]

A slotted ALOHA channel has $B = 2$ Mbps, frames of $L = 2000$ bits, and $n = 8$ nodes, each sending with $p = 0.1$. (a) Slot length? (b) Goodput? (c) Throughput (fraction of slots with any transmission)? (d) Useful bit rate? (e) Average time between successful frames?

Q2.8 ●●○ [R2 / own]

A channel carries on average $G = 0.4$ attempts per slot (Poisson model). (a) Slotted ALOHA: fraction of slots that are successful, empty, and collisions? (b) Pure ALOHA goodput? (c) Should the load be increased or decreased to maximize each?

Q2.9 ●●● [R2 Q1 variant]

Hosts C and D are both attached to two networks. The upper one runs slotted ALOHA and has 3 other nodes; the lower one runs pure ALOHA and has 2 other nodes. D never sends. Every other node sends in each of its slots with probability q. When C has a message for D it picks the upper network with probability p and then sends with probability q. (a) C has one attempt: probability that the message arrives? Numbers for $p = 0.6$, $q = 0.1$. (b) C now tries at the start of every slot (choosing a network each time). Goodput of the upper network? Same numbers.

Q2.10 ●●● [R2 Q2 variant]

One channel carries n slotted-ALOHA nodes (slots of length T, aligned to a common grid) and m pure-ALOHA nodes (frames of length T, own clocks, not aligned with the grid). Each node sends with probability p at the start of each of its slots. (a) Probability that a given slotted node succeeds in a slot? (b) That a given pure node succeeds? (c) Numbers for $n = m = 5$, $p = 0.1$. Who does better, and why?

CSMA/CD, Ethernet, WiFi

Q2.11 ●○○ [own]

A CSMA/CD LAN runs at 1 Gbps over at most 200 m of cable ($s = 2 \times 10^8$ m/s). What is the minimum frame size? Compare with Ethernet's 64 bytes.

Q2.12 ●○○ [KR 6.R6]

After the 5th collision, what is the probability that a node picks $K = 4$? How long is that delay on 10 Mbps Ethernet?

Q2.13 ●○○ [KR 6.P17]

For $K = 100$, how long does an adapter wait on a 100 Mbps channel? On a 1 Gbps channel?

Q2.14 ●●○ [KR 6.P18]

A and B are on a 10 Mbps channel, 325 bit times apart. A starts sending a minimum-size frame ($512 + 64 = 576$ bit times including the preamble) at $t = 0$; B starts before A's signal reaches it. Can A finish its frame before it hears B? What does that mean?

Q2.15 ●●○ [own]

Two NICs have just collided for the 2nd time. (a) From which set is K chosen? (b) Probability that they collide again? (c) Probability they collide on each of the next three attempts (3rd, 4th and 5th)?

Q2.16 ●●○ [lecture]

On a CSMA/CD LAN, PROP $= 2\,\mu$s and TRANSP $= 50\,\mu$s. (a) Efficiency in the simple model ($A = 1.5$) and the precise model? (b) The simple model with the large-N value $\alpha = 1/e$ instead of 0.4?

Q2.17 ●●● [KR 6.P20]

A CSMA/CD-like protocol: time is slotted with short slots; a frame lasts k slots. When the channel is free, each of N nodes transmits in a slot with probability p. If exactly one does, it holds the channel for its whole frame (k slots); otherwise the slot is wasted. (a) Efficiency for fixed N, p? (b) The best p? (c) The efficiency as $N \to \infty$? (d) What happens as $k \to \infty$?

Q2.18 ●○○ [own]

In 802.11, RTS and CTS take 25 μs each and a data frame takes 500 μs. (a) What is the reservation overhead per frame? (b) How much time does a collision waste with and without RTS/CTS? (c) When is RTS/CTS not worth it?

WEEK 02 · SOLUTIONS

Q2.1

Idea. Every packet carries 20 (TCP) + 20 (IP) + 14 (Ethernet) = 54 bytes of headers.

(a) Total $200 + 54 = 254$ bytes; useful 200/254.

(b) Add the 4-byte CRC and the 8-byte preamble: $254 + 12 = 266$ bytes; useful 200/266.

Answer. (a) 78.7%; (b) 75.2%.

Q2.2

Idea. Ask what each thing does: meaning for the user (7), data format or encryption (6), process-to-process delivery (4), finding a path between networks (3), sharing one link (2), the medium (1).

- HTTP: 7. TLS: 6. TCP ACKs: 4. IP routing: 3. CSMA/CD: 2 (the MAC part of layer 2). Fibre: 1.

Answer. as above. The Internet's 5-layer stack has no separate session (5) or presentation (6) layer: the application does that work itself, so TLS counts as "application".

Q2.3

Idea. TDMA gives every station a fixed 1/8 of the time, busy or not.

(a) $40/8 = 5$ Mbps per busy station.

(b) Used $3 \times 5 = 15$ Mbps; idle $40 - 15 = 25$ Mbps (5 of every 8 slots are empty).

(c) With $M = 3$ busy stations, the ideal is $R/M = 40/3 \approx 13.3$ Mbps each.

Answer. (a) 5 Mbps; (b) 25 Mbps wasted; (c) property 2 fails (R/M). Property 1 fails too: a station alone would still get only 5 Mbps, not 40.

Q2.4

Idea. Maximize by setting the derivative to zero (product rule), then let $N \to \infty$ using $(1 - \frac{1}{N})^N \to \frac{1}{e}$.

(a) Slotted, $f(p) = Np(1-p)^{N-1}$:

1. $f'(p) = N(1-p)^{N-1} + Np \cdot (N-1)(1-p)^{N-2} \cdot (-1)$.
2. Factor out $N(1-p)^{N-2}$: $f'(p) = N(1-p)^{N-2}[(1-p) - (N-1)p] = N(1-p)^{N-2}[1 - Np]$.
3. $f' = 0 \Rightarrow p^* = 1/N$ (the bracket is positive before and negative after, so it is a maximum).
4. $f(p^*) = N \cdot \frac{1}{N}(1 - \frac{1}{N})^{N-1} = (1 - \frac{1}{N})^{N-1} = \dfrac{(1 - \frac{1}{N})^N}{1 - \frac{1}{N}} \to \dfrac{1/e}{1}$.

(b) Pure, $g(p) = Np(1-p)^{2(N-1)}$:

1. In the same way: $g'(p) = N(1-p)^{2N-3}[(1-p) - 2(N-1)p] = N(1-p)^{2N-3}[1 - (2N-1)p]$, so $p^* = \frac{1}{2N-1}$.
2. With $M = 2N - 1$: $g(p^*) = \dfrac{N}{2N-1}\left(1 - \dfrac{1}{M}\right)^{M-1}$. The first factor $\to \frac{1}{2}$ and the second $\to \frac{1}{e}$.

Answer. (a) $p^* = 1/N$, limit $1/e \approx 0.37$; (b) $p^* = 1/(2N-1)$, limit $1/(2e) \approx 0.18$.

Q2.5

Idea. First find the per-slot probabilities. Slots are independent, so "first time in slot k" means fail $k-1$ times, then succeed (geometric).

With $p = 0.25$: $(1-p)^3 = 0.4219$.

(a) A wins one slot when A sends and B, C, D are silent: $q = p(1-p)^3 = 0.25 \times 0.4219 = 0.1055$. First time in slot 4: lose slots 1, 2, 3, win slot 4: $(1-q)^3 q = 0.8945^3 \times 0.1055 = 0.7157 \times 0.1055 = 0.0755$.

(b) "Some node succeeds" = exactly one of the four sends. The four events "node i alone" are disjoint, so add them: $s = 4p(1-p)^3 = 0.4219$.

(c) Same idea with s: $(1-s)^3 s = 0.5781^3 \times 0.4219 = 0.1932 \times 0.4219 = 0.0815$.

(d) Efficiency = the probability that a slot is successful = s.

Answer. (a) $(1-q)^3 q = 0.0755$; (b) $4p(1-p)^3 = 0.422$; (c) $(1-s)^3 s = 0.0815$; (d) 0.422, the best possible for 4 nodes, since $p = 1/4 = 1/N$.

Watch out: (a) is about one node (q, no factor 4); (b) and (c) are about any node (s, factor 4).

Q2.6

Idea. A slot is A's if A sends and B does not. Write each node's success probability and compare.

(a) A: $p_A(1-p_B)$. B: $p_B(1-p_A)$. These are disjoint, so the efficiency is $p_A(1-p_B) + p_B(1-p_A)$.

(b) Put $p_A = 2p_B$: A gets $2p_B(1-p_B)$ and B gets $p_B(1-2p_B)$. Their ratio is $\dfrac{2(1-p_B)}{1-2p_B}$, which is more than 2 because $1-p_B > 1-2p_B$. Reason: A sending more often also silences B more often. For exactly twice, solve $p_A(1-p_B) = 2p_B(1-p_A)$: $p_A - p_A p_B = 2p_B - 2p_A p_B \Rightarrow p_A(1+p_B) = 2p_B \Rightarrow p_A = \dfrac{2p_B}{1+p_B}$. Check with $p_B = 0.1$: $p_A = 0.182$; A gets $0.182 \times 0.9 = 0.164$, B gets $0.1 \times 0.818 = 0.082$, exactly half.

(c) A wins: A sends ($2p$) and the other $N-1$ are silent: $2p(1-p)^{N-1}$. Another node j wins: j sends (p), A is silent ($1-2p$), and the remaining $N-2$ are silent: $p(1-2p)(1-p)^{N-2}$.

Answer. (a) $p_A(1-p_B)$, total $p_A(1-p_B) + p_B(1-p_A)$; (b) no, more than twice; use $p_A = 2p_B/(1+p_B)$; (c) $2p(1-p)^{N-1}$ and $p(1-2p)(1-p)^{N-2}$.

Q2.7

Idea. Goodput = P(exactly one sends); throughput = P(at least one sends).

(a) $T = L/B = 2000/(2 \times 10^6) = 1\,\text{ms}$.

(b) $\eta = \binom{8}{1} 0.1 \cdot 0.9^7 = 0.8 \times 0.4783 = 0.383$.

(c) At least one sends $= 1 - P(\text{nobody}) = 1 - 0.9^8 = 1 - 0.4305 = 0.570$.

(d) $\eta \times B = 0.383 \times 2\,\text{Mbps} = 765\,\text{kbps}$.

(e) Successes are geometric with mean $1/\eta = 2.61$ slots, i.e. $2.61 \times 1\,\text{ms}$.

Answer. (a) 1 ms; (b) 0.383; (c) 0.570; (d) 765 kbps; (e) ≈ 2.6 ms. The channel is busy 57% of the time but useful only 38%.

Q2.8

Idea. Poisson model: the number of attempts in a slot is $X \sim \text{Pois}(G)$, so $P(X=k) = e^{-G}G^k/k!$.

(a) Success $P(X=1) = Ge^{-G} = 0.4 \times 0.6703 = 0.268$. Empty $P(X=0) = e^{-0.4} = 0.670$. Collision: the rest, $1 - 0.670 - 0.268 = 0.062$.

(b) A pure frame succeeds if no other frame starts in its $2T$ window: Pois($2G$) is 0 with probability e^{-2G}. So $\eta = Ge^{-2G} = 0.4\,e^{-0.8} = 0.4 \times 0.4493 = 0.180$.

(c) Slotted is best at $G^* = 1$ and pure at $G^* = 0.5$. $G = 0.4$ is below both.

Answer. (a) 0.268, 0.670, 0.062; (b) 0.180; (c) increase the load in both cases: up to $G = 1$ (slotted, max 0.368) and slightly, to $G = 0.5$ (pure, max 0.184).

Q2.9

Idea. (a) asks about *C's* message: split on the network C chooses (total probability). (b) asks about the *network*: a slot is useful if any one node succeeds.

(a) One attempt.

1. Upper (slotted): C sends (q) and each of the 3 others is silent in that one slot: $q(1-q)^3 = 0.1 \times 0.729 = 0.0729$.
2. Lower (pure): C's frame overlaps *two* slots of each other node, so each must skip both, $(1-q)^2$; with 2 nodes, $(1-q)^4$. So $q(1-q)^4 = 0.1 \times 0.6561 = 0.0656$.
3. Combine: $p \times 0.0729 + (1-p) \times 0.0656 = 0.6 \times 0.0729 + 0.4 \times 0.0656 = 0.0437 + 0.0262$.

(b) Goodput of the upper network.

1. C chose upper (p): 4 nodes, each sending with probability q; exactly one: $4q(1-q)^3 = 0.2916$.
2. C chose lower $(1-p)$: 3 nodes: $3q(1-q)^2 = 0.243$.
3. $\eta = 0.6 \times 0.2916 + 0.4 \times 0.243 = 0.175 + 0.097$.

Answer. (a) $q[p(1-q)^3 + (1-p)(1-q)^4] = 0.070$; (b) $p \cdot 4q(1-q)^3 + (1-p) \cdot 3q(1-q)^2 = 0.272$.

Watch out: one specific node (C) → no factor n; the whole network → factor n ($\binom{n}{1}$).

Q2.10

Idea. For each competitor, ask: how many of its slot starts fall in the window where a start would hit my frame? Each such start must be skipped.

(a) A slotted node's slot is $[t, t+T]$.

1. It sends: p.
2. The other slotted nodes use the same aligned slot; each must be silent: $(1-p)^{n-1}$.
3. A pure frame $[u, u+T]$ overlaps $[t, t+T]$ if $u \in (t-T,\ t+T)$, a window of length $2T$. A pure node starts every T, so exactly 2 of its starts fall there; it must skip both: $(1-p)^2$. With m pure nodes: $(1-p)^{2m}$.
4. Different nodes are independent, so multiply: $p(1-p)^{n-1}(1-p)^{2m}$.

(b) A pure node's frame $[u, u+T]$, with u between two grid points.

1. It sends: p.
2. The other pure nodes: 2 starts each in its window, as in plain pure ALOHA: $(1-p)^{2(m-1)}$.
3. The frame covers the end of one grid slot and the start of the next, i.e. **two** slots. Every slotted node must be silent in both: $(1-p)^{2n}$.
4. Multiply: $p(1-p)^{2(m-1)}(1-p)^{2n}$.

(c) Slotted: $0.1 \times 0.9^4 \times 0.9^{10} = 0.1 \times 0.9^{14} = 0.0229$. Pure: $0.1 \times 0.9^8 \times 0.9^{10} = 0.1 \times 0.9^{18} = 0.0150$.

Answer. (a) $p(1-p)^{n-1+2m}$; (b) $p(1-p)^{2(m-1)+2n}$; (c) 0.0229 vs. 0.0150. The slotted node does better: it is exposed to one slot of each slotted node, while the unaligned pure frame is exposed to two slots of everyone.

Q2.11

Idea. CSMA/CD only works if the sender is still transmitting when news of a collision comes back: TRANSP $\geq 2\,$PROP.

1. PROP $= d/s = 200/(2 \times 10^8) = 1\,\mu$s, so $2\,$PROP $= 2\,\mu$s.
2. TRANSP $= L/R \geq 2\,\mu$s gives $L \geq 2 \times 10^{-6} \times 10^9 = 2000$ bits $= 250$ bytes.

3. Ethernet's 64 bytes = 512 bits last only $512/10^9 = 0.512\,\mu$s $< 2\,\mu$s.

Answer. 250 bytes. 64 bytes is too short at 1 Gbps over 200 m: a collision could go unnoticed. (In practice gigabit Ethernet is switched, so there are no collisions.)

Q2.12

1. After the m-th collision, K is uniform on $\{0, \ldots, 2^m - 1\}$. Here $m = 5$: $\{0, \ldots, 31\}$, 32 values, so $P(K = 4) = 1/32$.

2. The wait is $K \cdot 512$ bit times = 2048 bit times. One bit at 10 Mbps takes $0.1\,\mu$s: $2048 \times 0.1 = 204.8\,\mu$s.

Answer. $1/32$; $204.8\,\mu$s.

Q2.13

1. $K \cdot 512 = 100 \times 512 = 51\,200$ bit times.

2. 100 Mbps: a bit time is 10 ns, so $51\,200 \times 10$ ns $= 512\,\mu$s. 1 Gbps: 1 ns, so $51.2\,\mu$s.

Answer. $512\,\mu$s and $51.2\,\mu$s.

Q2.14

Idea. Draw the timeline in bit times. The worst case is B starting at the last possible moment, just before A's signal reaches it.

1. $t = 0$: A starts. A's first bit reaches B at $t = 325$.

2. Just before $t = 325$, B still senses an idle line and starts sending.

3. B's first bit needs another 325 bit times to reach A: it arrives at $t \approx 650$.

4. A's minimum frame ends at $t = 576$, before 650.

Answer. Yes: A finishes at 576, before hearing B at 650, so A wrongly believes its frame got through. The cable is too long for this frame size (TRANSP $<$ 2 PROP).

Q2.15

Idea. Two nodes collide again exactly when they pick the same K. With k equally likely values, that happens with probability $k \cdot (1/k)^2 = 1/k$.

(a) After the 2nd collision: $\{0, 1, 2, 3\}$.

(b) 4 values: $1/4$.

(c) To collide on the next three attempts they must match from sets of 4, then 8 (after the 3rd collision), then 16 (after the 4th). The choices are independent: $\frac{1}{4} \cdot \frac{1}{8} \cdot \frac{1}{16}$.

Answer. (a) $\{0, \ldots, 3\}$; (b) $1/4$; (c) $1/512$. (This ignores the small effect of the busy-channel rule; it is the usual exam answer.)

Q2.16

Idea. $\eta = \dfrac{\text{TRANSP}}{\text{TRANSP} + A \cdot 2\,\text{PROP}}$, where A is the expected number of wasted 2 PROP slots per packet.

(a) $a = \text{PROP}/\text{TRANSP} = 2/50 = 0.04$. Simple model ($A = 1.5$): $\eta = 50/(50 + 1.5 \times 4) = 50/56 = 0.893$, which is the same as $1/(1 + 3a)$. Precise: $1/(1 + 5a) = 1/1.2 = 0.833$.

(b) $A = (1 - \alpha)/\alpha$ with $\alpha = 1/e$: $A = e - 1 = 1.718$. Wasted time $1.718 \times 4 = 6.87\,\mu$s, so $\eta = 50/56.87 = 0.879$.

Answer. (a) 0.893 (simple) and 0.833 (precise); (b) 0.879.

Q2.17

Idea. The channel alternates between contention (wasted slots) and a frame (k useful slots). Efficiency = useful ÷ total per cycle.

(a) A contention slot is won if exactly one node sends: $\alpha = Np(1-p)^{N-1}$. The number of wasted slots before a win is the number of failures before the first success of Geo(α): $x = (1-\alpha)/\alpha$. The winning slot is the first slot of the frame, so a cycle is x wasted $+\,k$ useful slots: efficiency $= \dfrac{k}{k + (1-\alpha)/\alpha}$.

(b) The efficiency grows as x shrinks, i.e. as α grows. α is maximized at $p^* = 1/N$ (Q2.4).

(c) $\alpha \to 1/e$, so $x \to (1 - \frac{1}{e})\, e = e - 1 \approx 1.72$.

(d) $\dfrac{k}{k + 1.72} \to 1$ as $k \to \infty$. For example $k = 100$: 0.983.

Answer. (a) $\dfrac{k}{k + (1-\alpha)/\alpha}$ with $\alpha = Np(1-p)^{N-1}$; (b) $p^* = 1/N$; (c) $\dfrac{k}{k + e - 1}$; (d) $\to 1$: long frames make contention negligible, the same as $a \to 0$ in the lecture.

Q2.18

(a) Extra time per frame: $25 + 25 = 50\,\mu$s on a $500\,\mu$s frame, i.e. 10%.

(b) Without RTS/CTS, a collision destroys a whole data frame: $500\,\mu$s. With it, only an RTS collides: $25\,\mu$s.

(c) It costs $50\,\mu$s on every frame and saves time only when there is a collision.

Answer. (a) 10%; (b) 500 vs. $25\,\mu$s; (c) not worth it when collisions are rare (light load, no hidden terminals) or frames are short.